```python
my_list=[10,3.5,"hello",True]
print("list:",my_list)
print("type of my_list:",type(my_list))
print("type of 10:",type(my_list[0]))
print("type of 3.5:", type(my_list[1]))
print("type of 'hello':", type(my_list[2]))
print("type of true:",type(my_list[3]))
```

✓ [1] 158ms

```
list: [10, 3.5, 'hello', True]
type of my_list: <class 'list'>
type of 10: <class 'int'>
type of 3.5: <class 'float'>
type of 'hello': <class 'str'>
type of true: <class 'bool'>
```

```python
a=[1,2,3]
b=[4,5,6]
print("concatenation:",a+b)
print("repetition:",a*3)
print("Is 2 in list a?",2 in a)
print("length of list a:",len(a))
```

✓ [2] 47ms

```
concatenation: [1, 2, 3, 4, 5, 6]
repetition: [1, 2, 3, 1, 2, 3, 1, 2, 3]
Is 2 in list a? True
length of list a: 3
```

```python
nums=[10,5,20,8,15]
print("maximum:",max(nums))
print("minimum:",min(nums))
print("sum:",sum(nums))
nums.sort()
print("ascending order:",nums)
nums.sort(reverse=True)
print("descending order:",nums)
print("index of 8:",nums.index(8))
```

✓ [3] 58ms

```
maximum: 20
minimum: 5
sum: 58
ascending order: [5, 8, 10, 15, 20]
descending order: [20, 15, 10, 8, 5]
index of 8: 3
```

```python
my_list=[10,20,30]
my_list.append(40)
print("after append:",my_list)
my_list.insert(1,15)
print("after insert:",my_list)
my_list.remove(20)
print("after remove:",my_list)
my_list.pop()
print("after pop:",my_list)
my_list.reverse()
print("after reverse:",my_list)
```

✓ [4] 60ms

```
after append: [10, 20, 30, 40]
after insert: [10, 15, 20, 30, 40]
after remove: [10, 15, 30, 40]
after pop: [10, 15, 30]
after reverse: [30, 15, 10]
```

```python
numbers=list(map(int,input("enter numbers seperated by space:").split()))
even_nums=[num for num in numbers if num%2==0]
odd_nums=[num for num in numbers if num%2!=0]
largest=max(numbers)
smallest=min(numbers)
unique_numbers=list(set(numbers))
print("even numbers:",even_nums)
print("odd numbers:",odd_nums)
print("largest:",largest)
print("smallest:",smallest)
print("list without duplicates:",unique_numbers)
```

✓ [6] 20s 145ms

```
even numbers: [10, 8, 10, 2]
odd numbers: [5, 3, 5]
largest: 10
smallest: 2
list without duplicates: [2, 3, 5, 8, 10]
```